Md. Yousuf Hossain

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Portfolio — Facebook

Professional Summary

Aspiring Cybersecurity Specialist and Computer Science undergraduate with a strong foundation in network security and ethical hacking. Proven track record in Capture The Flag (CTF) competitions and hands-on experience with security tools like Hack The Box and TryHackMe. Passionate about identifying vulnerabilities and researching secure transmission mechanisms. Eager to leverage technical skills in a professional environment to contribute to robust security solutions and innovative research.

Education

Bachelor of Science in Computer Science and Engineering (CSE) 2023 – Present
Dhaka International University Dhaka, Bangladesh
5th Semester, 3rd Year (Expected Graduation: 2027)
Future Major: Application Development, App Bug Bounty, Cybersecurity.

Higher Secondary Certificate (HSC) 2022
GPA: 4.67 / 5.00

Technical Skills

- **Programming Languages:** Python, Flutter (Dart), C, Java, SQL
- **Cybersecurity & Ethical Hacking:** CTF Basics, Linux Fundamentals, Ethical Hacking Principles, Network Security
- **Tools & Platforms:** Hack The Box, TryHackMe, PicoCTF, Git, GitHub, VS Code, Android Studio

Featured Projects

Visiolock: Secure Image Transmission *Developer*

- Developed a cross-media secure image transmission system utilizing audio-domain encoding and device-bound cryptography.
- Focused on enhancing data privacy and security during transmissions through non-traditional communication channels.

Secure NFC: Mobile Ticketing System *Researcher*

- Investigated NFC-based mobile ticketing frameworks using Host Card Emulation (HCE).
- Implemented and analyzed security protocols to mitigate Replay Attacks and ensure transactional integrity.

Sudoku Solver *Developer*

- Engineered a Java-based application implementing OOP principles and backtracking algorithms to solve complex Sudoku puzzles.
- Designed an interactive UI to visualize the step-by-step backtracking process.

University Database Management System *Developer*

- architected a comprehensive database solution to manage university informatics, including student records, academic cycles, and department logistics.
- Optimized SQL queries for efficient data handling and relational management.

Hostel Management System *Coder & Planner*

- Developed a C-based management tool for room allocation and student lifecycle tracking in a collaborative group environment.

Extracurricular Activities

- **Treasurer**, DIU Computer Programming Club (DIU CPC) 2025 – 2026
- **Executive Member**, BASIS (Bangladesh Association of Software and Information Services)
- **Organizer**, CSE Fest Fall 2024 (Successfully coordinated technical events and student logistics)
- **Volunteer**, 3rd BIM 2025
- **Executive Member**, DIU Computer Programming Club (DIU CPC) 2024 – 2025

Publications

1. **Secure NFC-Based Mobile Ticketing System Using Host Card Emulation (HCE):** Implementation and Analysis of Security Mechanisms Against Replay Attacks. *[Accepted]*
2. **A Real-Time Lightweight Intrusion Detection System for Smart City Networks Using Optimized Deep Learning.** *[Under Review]*
3. **AI-Driven Real-Time Reading Assessment and Assistance System for Dyslexia Using Speech and Behavioral Analytics.** *[Under Review]*
4. **VisioLock: Cross-Media Secure Image Transmission via Audio-Domain Encoding and Device-Bound Cryptography.** *[Under Review]*
5. **A Multimodal, Infrastructure-Free Offline Emergency Communication System for Mobile Devices Using Adaptive Hybrid Mesh Networking.** *[Under Review]*
6. **SafeLink+: A Zero-Trust Mobile Link Execution Framework for Protecting Users from Malicious URLs.** *[Under Review]*

I hereby affirm that the information provided above is accurate and true to the best of my knowledge.

Signature